

WL and SL pics: MLMs

Multilevel models

Est dist

```
#Sanity check: Stim distance and estimated distance with random int for participants
Null.dist <- lmer(subject_estimate ~ 1 + stim_distance + (1|subject), data = DdataSW)
coef(Null.dist)
```

```
## $subject
##               (Intercept) stim_distance
## 03b16361e0733153a514 -46.689586      0.8756043
## 03c20eaf3c5546c146e3 -18.021206      0.8756043
## 07c208d8989c09cafdcb  31.213100      0.8756043
## 08c2c1e4288f21b62161 -32.142687      0.8756043
## 0bc19d73dfcc79108774 323.826144      0.8756043
## 0d592ffa0ef1a0d4a525 204.382157      0.8756043
## 0f2434b6a8611d7299c7 211.074501      0.8756043
## 1153370ec05190a85eb0 -30.370116      0.8756043
## 19100d3830890f297f5c   5.527849      0.8756043
## 1a4a35d6d316c76b2554 -43.298067      0.8756043
## 1b12b81d3437180234b3 -29.152951      0.8756043
## 1d3cd1067f2c7d38057d 122.368423      0.8756043
## 20616fb59a3d687092d0 -24.610442      0.8756043
## 23f4a5027e9b70896a82 -39.587485      0.8756043
## 292b6a1ff66453541947 -39.894731      0.8756043
## 2ab98650a9465bb38516   8.212844      0.8756043
## 2c0ad1a3bae34df22e2d -11.592682      0.8756043
## 2c7549269a6dc4bf5c72  57.785744      0.8756043
## 2d6a9afc22d6a48697de 103.914384      0.8756043
## 38e4849b03aad5089e0e -29.720174      0.8756043
## 39cb033ba23e0a4204b4 -28.703900      0.8756043
## 3a4ced6c4fc52ad4271 -32.639007      0.8756043
## 3d85f7a3c49d9a89a36d 171.846664      0.8756043
## 41956ce17cf84a1253c9  -8.626580      0.8756043
## 499d33b349396d9f4919 -37.105886      0.8756043
## 4d0ec1b598470d393c99 148.319212      0.8756043
## 575408d27d4f9024a6a3 -72.652314      0.8756043
## 58024b3726839b0ad500 -47.764945      0.8756043
## 584eecb0fbfc50073f14 -19.616519      0.8756043
## 5b06e8406e2b0e210a97 192.157187      0.8756043
## 5ce421047cadb9563298 245.995588      0.8756043
## 635517b9b32b3fa70f83 -16.047743      0.8756043
## 64e6dc024b446b55739a -41.631850      0.8756043
## 6bcd49af023f477bbf37  64.185907      0.8756043
## 6bfa6a31a326dc706fdc -55.268829      0.8756043
## 6c5aa99681eb4d6905b1 271.634425      0.8756043
## 736bdf214487177022e5 -40.769199      0.8756043
```

```

## 7543eafb309fd95fad6 13.270580 0.8756043
## 7fd7243a168e383f9e8b 160.191749 0.8756043
## 826e1d16e3b533fb245a -26.151605 0.8756043
## 83538a13d227275b86a5 -34.683372 0.8756043
## 875b5d029de6eaeab7e7 11.356203 0.8756043
## 87b67b2c3ce58ae2ae1e -19.143834 0.8756043
## 8de5b549a2ea6dc6e183 -62.477757 0.8756043
## 8f80bcc354d05f533ab4 509.226023 0.8756043
## 945fe99a248b31456606 350.200847 0.8756043
## 97d9363ed595442930a1 -31.483291 0.8756043
## 98b0f6a6c10bbac3ed08 -25.466003 0.8756043
## 9cc3e4db792b9a5f4e79 -12.857116 0.8756043
## 9e57b44f8d0fde900aac -9.524682 0.8756043
## a07e6c06f538e78a6fc5 -30.322848 0.8756043
## a4809ff054019f364d68 69.458707 0.8756043
## a99e74b610c40ecaa76f -25.997775 0.8756043
## b032e098afde0db1fe47 -18.883857 0.8756043
## b6af4fba0707225b88c6 -15.090555 0.8756043
## b759a10905ff6acf8c8b -48.549604 0.8756043
## b849ace4da81775ebd98 7.338376 0.8756043
## b97dd7e424b184fe9dd7 -72.671474 0.8756043
## ba912daea85edd41c794 -1.240868 0.8756043
## bb6bb1c0eb581072ba99 150.335659 0.8756043
## bd47d8131e7c18207855 38.748332 0.8756043
## c13597eae01d4a01c6b9 75.798846 0.8756043
## c25a28933629f381f987 141.398848 0.8756043
## c2e99bdb4d91daa8dda5 15.409482 0.8756043
## c309013be7be5bb5e89c 265.952260 0.8756043
## c34b33f5799854f89af7 -24.579718 0.8756043
## cbc292b667a3879ecd4a 22.452497 0.8756043
## cc90d814885eaf54262f -35.309681 0.8756043
## d2d7e4c2d26d70b0994c 30.346346 0.8756043
## d34f86c865d6a3244c0d -37.224057 0.8756043
## db6df130100a85567a03 -22.015399 0.8756043
## dc0a2924029e00ade90d 50.100822 0.8756043
## ddc74e6981d96523a1d4 -60.447572 0.8756043
## ddd964c9902c136806eb -27.628540 0.8756043
## e1e8d661d37cf51a1b8c -45.342432 0.8756043
## e338df513c9d760904b9 -28.420288 0.8756043
## eb5209c2c5d5ef9d8913 -67.239591 0.8756043
## ed1fd295ccf29bd62cb1 -26.565052 0.8756043
## ed98406eb2fb2cee3586 279.488180 0.8756043
## f1d2903ba93367937b48 -9.394694 0.8756043
## f4146d4a6503f38cd7b4 -17.962120 0.8756043
## f559a201d79bdf600dad -68.277136 0.8756043
## f705c2504ed873931a28 -43.841655 0.8756043
## f7f047a44e75f386525c -33.017155 0.8756043
## f8d7b5b6a91e756d54ad 46.618547 0.8756043
## fa18c06cc119aa932a87 358.880092 0.8756043
##
## attr(,"class")
## [1] "coef.mer"

```

```
summary(Null.dist)
```

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method ['lmerModLmerTest']
## Formula: subject_estimate ~ 1 + stim_distance + (1 | subject)
## Data: DdataSW
##
## REML criterion at convergence: 82912.9
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -7.7134 -0.4278  0.0424  0.3442  7.4143
##
## Random effects:
##  Groups   Name                Variance Std.Dev.
## subject  (Intercept) 13764      117.32
## Residual                    8573       92.59
## Number of obs: 6936, groups:  subject, 86
##
## Fixed effects:
##              Estimate Std. Error      df t value Pr(>|t|)
## (Intercept)  3.525e+01  1.280e+01  8.728e+01  2.754  0.00716 **
## stim_distance 8.756e-01  1.788e-02  6.852e+03  48.973 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##              (Intr)
## stim_distnc -0.121
```

```
#Convergence probs with stimuli distance as a random effect
#7 unique stim distance
unique(Ddata$stim_distance)
```

```
## [1] 15 25 50 75 100 150 200
```

```
Null.dist2 <- lmer(subject_estimate ~ 1 + stim_distance + (1 + stim_distance|subject), data = DdataSW)
```

```
## Warning in checkConv(attr(opt, "derivs"), opt$par, ctrl = control$checkConv, : Model failed to
## converge with max|grad| = 0.00690089 (tol = 0.002, component 1)
```

```
coef(Null.dist2)
```

```
## $subject
##              (Intercept) stim_distance
## 03b16361e0733153a514   -5.0556995    0.398634406
## 03c20eaf3c5546c146e3    26.9364440    0.359916037
## 07c208d8989c09cafdcb    84.6748323    0.267328742
## 08c2c1e4288f21b62161    34.6165005    0.109904512
## 0bc19d73dfcc79108774   169.7187537    2.471733153
## 0d592ffa0ef1a0d4a525  -115.6501169    4.664312451
## 0f2434b6a8611d7299c7   -26.6557901    3.804224022
## 1153370ec05190a85eb0     4.1269657    0.480247745
## 19100d3830890f297f5c    13.3138562    0.786648342
## 1a4a35d6d316c76b2554   -1.5262843    0.396984993
## 1b12b81d3437180234b3    27.8823502    0.221478952
## 1d3cd1067f2c7d38057d    98.4145497    2.262130382
```

## 20616fb59a3d687092d0	15.2660161	0.418378376
## 23f4a5027e9b70896a82	16.6234686	0.231146750
## 292b6a1ff66453541947	7.4356505	0.333104188
## 2ab98650a9465bb38516	9.2966545	0.863086804
## 2c0ad1a3bae34df22e2d	42.4724615	0.255233772
## 2c7549269a6dc4bf5c72	90.6438559	0.497339570
## 2d6a9afc22d6a48697de	14.9168260	1.875329734
## 38e4849b03aad5089e0e	16.4275162	0.346482834
## 39cb033ba23e0a4204b4	21.2143124	0.303176321
## 3a4cede6c4fc52ad4271	30.0371832	0.156788159
## 3d85f7a3c49d9a89a36d	-47.5427892	3.390980791
## 41956ce17cf84a1253c9	49.8555833	0.204467157
## 499d33b349396d9f4919	26.9381964	0.141171737
## 4d0ec1b598470d393c99	78.6684078	1.672399791
## 575408d27d4f9024a6a3	1.2062346	0.029194106
## 58024b3726839b0ad500	13.6851562	0.171159802
## 584eecb0fbfc50073f14	44.5705612	0.139188047
## 5b06e8406e2b0e210a97	18.3646254	7.788567104
## 5ce421047cadb9563298	-59.1895051	5.307831141
## 635517b9b32b3fa70f83	-5.6756373	0.756929050
## 64e6dc024b446b55739a	23.8422839	0.124842856
## 6bcd49af023f477bbf37	28.9536140	1.278911087
## 6bfa6a31a326dc706fdc	-2.6073518	0.272202390
## 6c5aa99681eb4d6905b1	-77.4278218	4.843870169
## 736bdf214487177022e5	1.1811509	0.394885536
## 7543eafb309fd95fad6	-13.1486759	1.178730700
## 7fd7243a168e383f9e8b	17.6108554	2.475520888
## 826e1d16e3b533fb245a	49.3048402	0.018418349
## 83538a13d227275b86a5	21.1950159	0.234868789
## 875b5d029de6eaeab7e7	45.7409966	0.480720835
## 87b67b2c3ce58ae2ae1e	30.9210184	0.301305892
## 8de5b549a2ea6dc6e183	11.8584910	0.023511008
## 8f80bcc354d05f533ab4	456.1352784	1.270446280
## 945fe99a248b31456606	79.3957317	4.560794799
## 97d9363ed595442930a1	15.2372265	0.339941096
## 98b0f6a6c10bbac3ed08	31.7617989	0.219196882
## 9cc3e4db792b9a5f4e79	2.4292039	0.700450123
## 9e57b44f8d0fde900aac	-1.8906304	0.788235102
## a07e6c06f538e78a6fc5	-2.7689530	0.559956532
## a4809ff054019f364d68	145.8500566	-0.010313793
## a99e74b610c40ecaa76f	29.0294927	0.244470031
## b032e098afde0db1fe47	27.5696058	0.342760561
## b6af4fba0707225b88c6	29.4778482	0.364327380
## b759a10905ff6acf8c8b	18.7592453	0.103915109
## b849ace4da81775ebd98	-6.1201854	1.030054354
## b97dd7e424b184fe9dd7	4.8053320	-0.004832205
## ba912daea85edd41c794	50.4236769	0.282590867
## bb6bb1c0eb581072ba99	118.0746490	1.208438481
## bd47d8131e7c18207855	83.0603437	0.366217939
## c13597eae01d4a01c6b9	-14.4053142	1.909776226
## c25a28933629f381f987	-44.2631659	2.993573290
## c2e99bdb4d91daa8dda5	-15.2074987	1.226879916
## c309013be7be5bb5e89c	125.0850363	2.520672997
## c34b33f5799854f89af7	-9.7526201	0.705951365

```
## cbc292b667a3879ecd4a 22.5305433 0.874354780
## cc90d814885eaf54262f 11.2878057 0.341428357
## d2d7e4c2d26d70b0994c -29.0097012 1.556520242
## d34f86c865d6a3244c0d 6.0061104 0.380123545
## db6df130100a85567a03 26.5313733 0.318790030
## dc0a2924029e00ade90d 42.0896342 0.966680723
## ddc74e6981d96523a1d4 0.2452891 0.180101156
## ddd964c9902c136806eb 40.0816826 0.098898089
## e1e8d661d37cf51a1b8c 1.5525682 0.338209019
## e338df513c9d760904b9 31.7665030 0.185284644
## eb5209c2c5d5ef9d8913 -0.2805626 0.108296648
## ed1fd295ccf29bd62cb1 51.3889378 -0.020299947
## ed98406eb2fb2cee3586 88.2497627 3.184633383
## f1d2903ba93367937b48 37.3590424 0.339127762
## f4146d4a6503f38cd7b4 12.8160316 0.522699442
## f559a201d79bdf600dad 2.9249776 0.059605149
## f705c2504ed873931a28 9.3358743 0.266054494
## f7f047a44e75f386525c 27.7542881 0.178662573
## f8d7b5b6a91e756d54ad -17.1527131 1.606889884
## fa18c06cc119aa932a87 142.1601681 3.322106351
##
## attr(,"class")
## [1] "coef.mer"
```

```
fixef(Null.dist2)
```

```
## (Intercept) stim_distance
## 28.648364 1.037965
```

```
ranef(Null.dist2)
```

```
## $subject
## (Intercept) stim_distance
## 03b16361e0733153a514 -33.7040638 -0.639330212
## 03c20eaf3c5546c146e3 -1.7119203 -0.678048581
## 07c208d8989c09cafdcb 56.0264680 -0.770635876
## 08c2c1e4288f21b62161 5.9681362 -0.928060106
## 0bc19d73dfcc79108774 141.0703894 1.433768535
## 0d592ffa0ef1a0d4a525 -144.2984812 3.626347833
## 0f2434b6a8611d7299c7 -55.3041544 2.766259404
## 1153370ec05190a85eb0 -24.5213986 -0.557716872
## 19100d3830890f297f5c -15.3345081 -0.251316275
## 1a4a35d6d316c76b2554 -30.1746486 -0.640979624
## 1b12b81d3437180234b3 -0.7660141 -0.816485666
## 1d3cd1067f2c7d38057d 69.7661854 1.224165764
## 20616fb59a3d687092d0 -13.3823483 -0.619586242
## 23f4a5027e9b70896a82 -12.0248957 -0.806817868
## 292b6a1ff66453541947 -21.2127138 -0.704860430
## 2ab98650a9465bb38516 -19.3517098 -0.174877814
## 2c0ad1a3bae34df22e2d 13.8240972 -0.782730846
## 2c7549269a6dc4bf5c72 61.9954916 -0.540625048
## 2d6a9afc22d6a48697de -13.7315383 0.837365116
## 38e4849b03aad5089e0e -12.2208481 -0.691481783
## 39cb033ba23e0a4204b4 -7.4340519 -0.734788297
## 3a4ced6c4fc52ad4271 1.3888189 -0.881176459
## 3d85f7a3c49d9a89a36d -76.1911535 2.353016173
```

## 41956ce17cf84a1253c9	21.2072190	-0.833497460
## 499d33b349396d9f4919	-1.7101679	-0.896792881
## 4d0ec1b598470d393c99	50.0200435	0.634435173
## 575408d27d4f9024a6a3	-27.4421297	-1.008770512
## 58024b3726839b0ad500	-14.9632081	-0.866804816
## 584eecb0fbfc50073f14	15.9221969	-0.898776570
## 5b06e8406e2b0e210a97	-10.2837389	6.750602486
## 5ce421047cadb9563298	-87.8378694	4.269866523
## 635517b9b32b3fa70f83	-34.3240016	-0.281035568
## 64e6dc024b446b55739a	-4.8060804	-0.913121762
## 6bcd49af023f477bbf37	0.3052497	0.240946469
## 6bfa6a31a326dc706fdc	-31.2557161	-0.765762228
## 6c5aa99681eb4d6905b1	-106.0761861	3.805905551
## 736bdf214487177022e5	-27.4672134	-0.643079081
## 7543eafb309fd95fad6	-41.7970402	0.140766082
## 7fd7243a168e383f9e8b	-11.0375089	1.437556270
## 826e1d16e3b533fb245a	20.6564759	-1.019546269
## 83538a13d227275b86a5	-7.4533485	-0.803095829
## 875b5d029de6eaeab7e7	17.0926323	-0.557243783
## 87b67b2c3ce58ae2ae1e	2.2726541	-0.736658725
## 8de5b549a2ea6dc6e183	-16.7898733	-1.014453610
## 8f80bcc354d05f533ab4	427.4869141	0.232481662
## 945fe99a248b31456606	50.7473674	3.522830181
## 97d9363ed595442930a1	-13.4111378	-0.698023521
## 98b0f6a6c10bbac3ed08	3.1134346	-0.818767736
## 9cc3e4db792b9a5f4e79	-26.2191604	-0.337514495
## 9e57b44f8d0fde900aac	-30.5389948	-0.249729515
## a07e6c06f538e78a6fc5	-31.4173173	-0.478008086
## a4809ff054019f364d68	117.2016923	-1.048278410
## a99e74b610c40ecaa76f	0.3811284	-0.793494587
## b032e098afde0db1fe47	-1.0787585	-0.695204057
## b6af4fba0707225b88c6	0.8294839	-0.673637237
## b759a10905ff6acf8c8b	-9.8891190	-0.934049509
## b849ace4da81775ebd98	-34.7685497	-0.007910264
## b97dd7e424b184fe9dd7	-23.8430323	-1.042796823
## ba912daea85edd41c794	21.7753126	-0.755373751
## bb6bb1c0eb581072ba99	89.4262847	0.170473863
## bd47d8131e7c18207855	54.4119794	-0.671746679
## c13597eae01d4a01c6b9	-43.0536785	0.871811608
## c25a28933629f381f987	-72.9115302	1.955608672
## c2e99bdb4d91daa8dda5	-43.8558630	0.188915298
## c309013be7be5bb5e89c	96.4366720	1.482708380
## c34b33f5799854f89af7	-38.4009844	-0.332013253
## cbc292b667a3879ecd4a	-6.1178210	-0.163609837
## cc90d814885eaf54262f	-17.3605586	-0.696536261
## d2d7e4c2d26d70b0994c	-57.6580655	0.518555624
## d34f86c865d6a3244c0d	-22.6422539	-0.657841073
## db6df130100a85567a03	-2.1169910	-0.719174588
## dc0a2924029e00ade90d	13.4412699	-0.071283894
## ddc74e6981d96523a1d4	-28.4030752	-0.857863461
## ddd964c9902c136806eb	11.4333183	-0.939066529
## e1e8d661d37cf51a1b8c	-27.0957961	-0.699755599
## e338df513c9d760904b9	3.1181387	-0.852679973
## eb5209c2c5d5ef9d8913	-28.9289269	-0.929667969

```
## ed1fd295ccf29bd62cb1    22.7405735   -1.058264565
## ed98406eb2fb2cee3586    59.6013983    2.146668765
## f1d2903ba93367937b48     8.7106781   -0.698836855
## f4146d4a6503f38cd7b4   -15.8323327   -0.515265176
## f559a201d79bdf600dad   -25.7233867   -0.978359469
## f705c2504ed873931a28   -19.3124900   -0.771910124
## f7f047a44e75f386525c    -0.8940762   -0.859302045
## f8d7b5b6a91e756d54ad   -45.8010775    0.568925266
## fa18c06cc119aa932a87   113.5118038    2.284141733
##
## with conditional variances for "subject"
Null.cond <- lmer(subject_estimate ~ 1 + condition + (1|subject),
                  data = DdataSW)
summary(Null.cond)

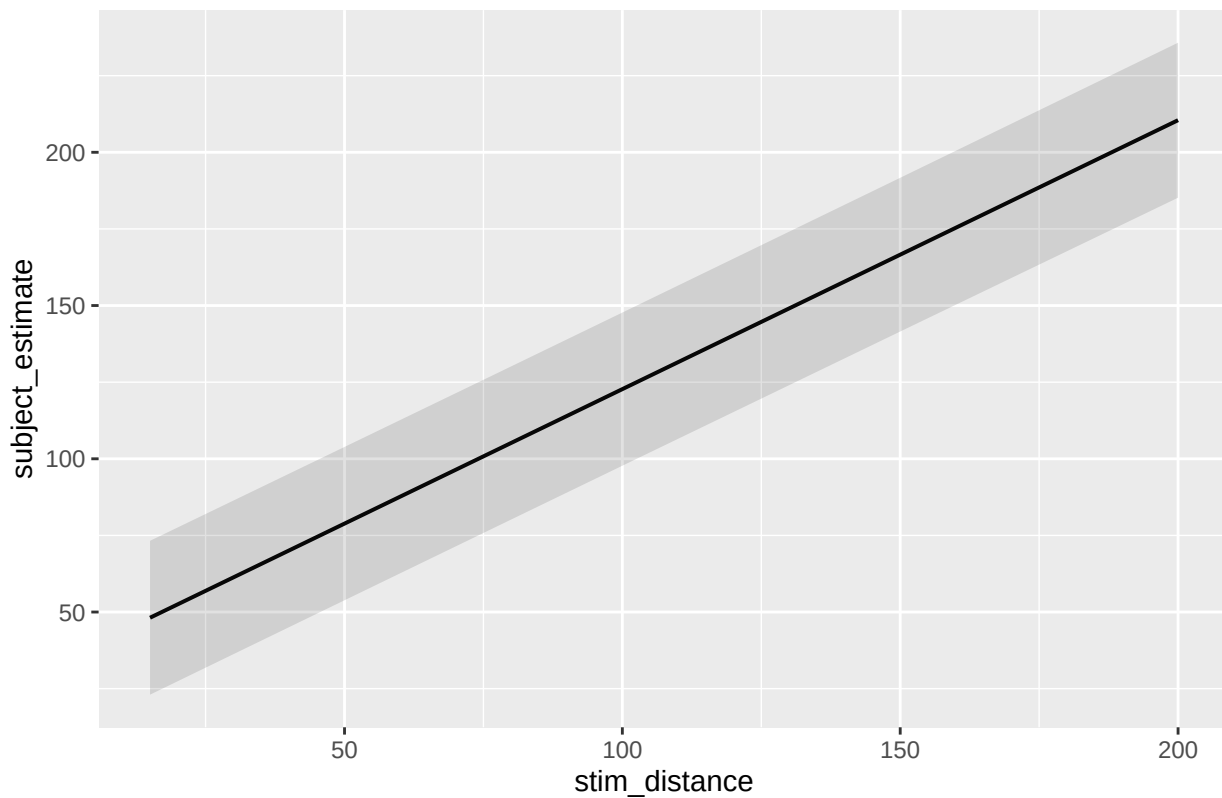
## Linear mixed model fit by REML. t-tests use Satterthwaite's method ['lmerModLmerTest']
## Formula: subject_estimate ~ 1 + condition + (1 | subject)
## Data: DdataSW
##
## REML criterion at convergence: 84955.1
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -5.6729 -0.2889 -0.0339  0.1992  7.3201
##
## Random effects:
## Groups Name Variance Std.Dev.
## subject (Intercept) 13956 118.1
## Residual 11573 107.6
## Number of obs: 6936, groups: subject, 86
##
## Fixed effects:
##              Estimate Std. Error    df t value Pr(>|t|)
## (Intercept)  109.110    40.810  83.783   2.674  0.00902 **
## condition      1.302    25.628  83.736   0.051  0.95959
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##              (Intr)
## condition -0.949

#Stim distance and condition
Modell1.dist.cond <- lmer(subject_estimate ~ 1 + stim_distance*condition +
                          (1|subject), data = DdataSW)
summary(Modell1.dist.cond)

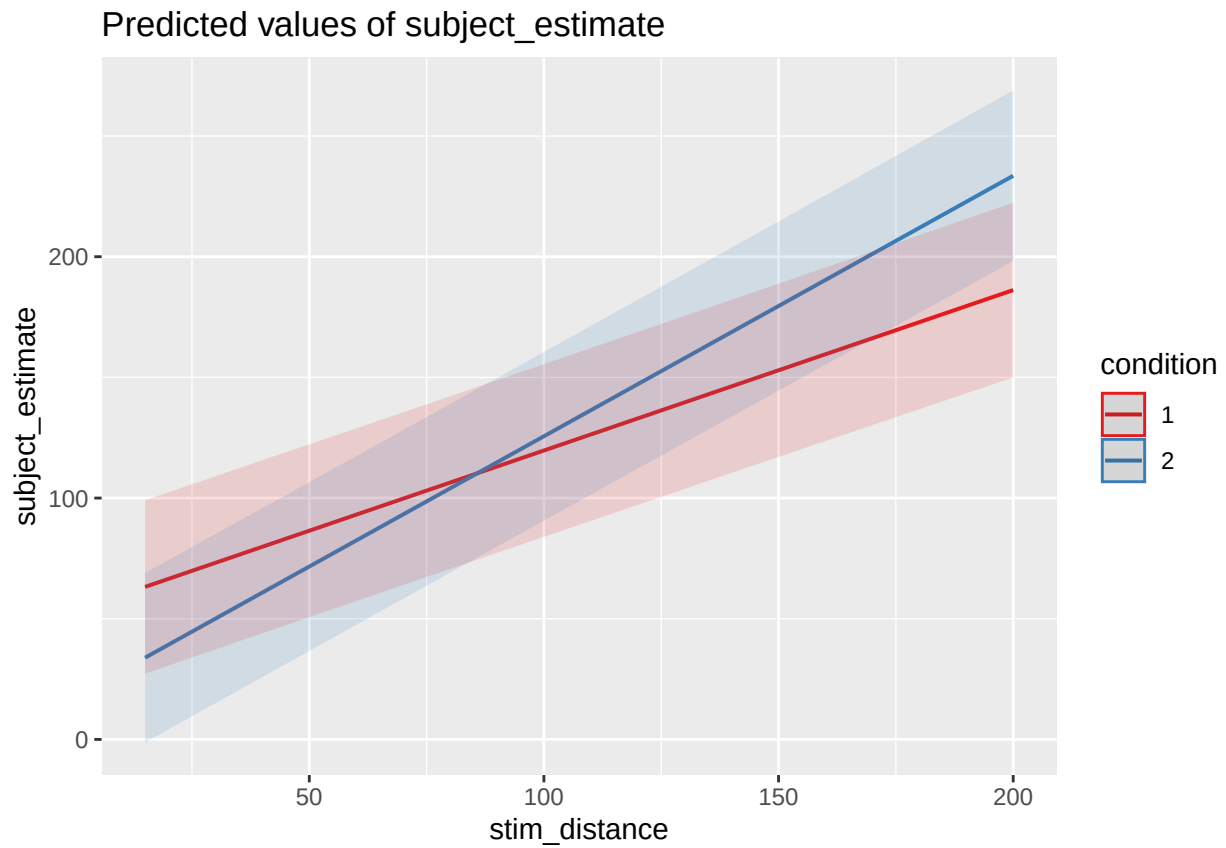
## Linear mixed model fit by REML. t-tests use Satterthwaite's method ['lmerModLmerTest']
## Formula: subject_estimate ~ 1 + stim_distance * condition + (1 | subject)
## Data: DdataSW
##
## REML criterion at convergence: 82773.6
##
## Scaled residuals:
```

```
##      Min      1Q  Median      3Q      Max
## -8.0525 -0.4069  0.0450  0.3106  7.2316
##
## Random effects:
## Groups   Name      Variance Std.Dev.
## subject  (Intercept) 13860    117.73
## Residual                8407     91.69
## Number of obs: 6936, groups: subject, 86
##
## Fixed effects:
##              Estimate Std. Error      df t value Pr(>|t|)
## (Intercept)    88.79307   40.89587   86.21947   2.171   0.0327 *
## stim_distance     0.25003    0.05628 6850.85238   4.443 9.03e-06 ***
## condition     -35.56944   25.68534   86.20909  -1.385   0.1697
## stim_distance:condition  0.41470    0.03541 6851.08155  11.710 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##      (Intr) stm_ds condtn
## stim_distnc -0.119
## condition   -0.949  0.113
## stm_dstnc:c  0.113 -0.949 -0.119
plot_model(Model1.dist.cond, type = "pred", terms = "stim_distance")
```

Predicted values of subject_estimate




```
plot_model(Model1.dist.cond, type = "int", terms = "condition")
```



```
tab_model(Model1.dist.cond, show.df = TRUE)
```

subject_estimate

Predictors

Estimates

CI

p

df

(Intercept)

88.79

8.62 – 168.96

0.030

6930.00

stim distance

0.25

0.14 – 0.36

<0.001
 6930.00
 condition
 -35.57
 -85.92 – 14.78
 0.166
 6930.00
 stim distance * condition
 0.41
 0.35 – 0.48
 <0.001
 6930.00
 Random Effects
 2
 8406.60
 00 subject
 13860.11
 ICC
 0.62
 N subject
 86
 Observations
 6936
 Marginal R2 / Conditional R2
 0.125 / 0.670

#Model comparison

```
anova(Null.dist, Null.cond, Model1.dist.cond)
```

```
## refitting model(s) with ML (instead of REML)
```

```
## Data: DdataSW
```

```
## Models:
```

```
## Null.dist: subject_estimate ~ 1 + stim_distance + (1 | subject)
```

```
## Null.cond: subject_estimate ~ 1 + condition + (1 | subject)
```

```
## Model1.dist.cond: subject_estimate ~ 1 + stim_distance * condition + (1 | subject)
```

```
##          npar    AIC    BIC logLik deviance  Chisq Df Pr(>Chisq)
```

```
## Null.dist          4 82922 82949 -41457    82914
```

```
## Null.cond          4 84978 85006 -42485    84970    0.0  0
```

```
## Model1.dist.cond    6 82790 82831 -41389    82778 2192.6  2 < 2.2e-16 ***
```

```
## ---
```

```
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Ratio

```
#Sanity check: Stim distance and estimated distance with random int for participants
Null.distr <- lmer(Ratio ~ 1 + stim_distance + (1|subject), data = DdataSW)
coef(Null.distr)
```

```
## $subject
##               (Intercept) stim_distance
## 03b16361e0733153a514    0.6976788 -0.004572989
## 03c20eaf3c5546c146e3    1.2749770 -0.004572989
## 07c208d8989c09cafdcb    2.4330031 -0.004572989
## 08c2c1e4288f21b62161    1.2391279 -0.004572989
## 0bc19d73dfcc79108774    5.3263884 -0.004572989
## 0d592ffa0ef1a0d4a525    3.0261260 -0.004572989
## 0f2434b6a8611d7299c7    3.5096160 -0.004572989
## 1153370ec05190a85eb0    0.9078155 -0.004572989
## 19100d3830890f297f5c    1.5016416 -0.004572989
## 1a4a35d6d316c76b2554    0.7553138 -0.004572989
## 1b12b81d3437180234b3    1.1207766 -0.004572989
## 1d3cd1067f2c7d38057d    8.1731594 -0.004572989
## 20616fb59a3d687092d0    1.1328093 -0.004572989
## 23f4a5027e9b70896a82    1.0129727 -0.004572989
## 292b6a1ff66453541947    0.8924488 -0.004572989
## 2ab98650a9465bb38516    1.4281502 -0.004572989
## 2c0ad1a3bae34df22e2d    1.5414451 -0.004572989
## 2c7549269a6dc4bf5c72    2.6796081 -0.004572989
## 2d6a9afc22d6a48697de    2.5596642 -0.004572989
## 38e4849b03aad5089e0e    1.1049161 -0.004572989
## 39cb033ba23e0a4204b4    1.0812340 -0.004572989
## 3a4cede6c4fc52ad4271    1.2726463 -0.004572989
## 3d85f7a3c49d9a89a36d    2.9646767 -0.004572989
## 41956ce17cf84a1253c9    1.6649712 -0.004572989
## 499d33b349396d9f4919    1.1014596 -0.004572989
## 4d0ec1b598470d393c99    3.4023646 -0.004572989
## 575408d27d4f9024a6a3    0.4549252 -0.004572989
## 58024b3726839b0ad500    0.8396726 -0.004572989
## 584eecb0fbfc50073f14    1.5002433 -0.004572989
## 5b06e8406e2b0e210a97    8.2406454 -0.004572989
## 5ce421047cadb9563298    4.1420491 -0.004572989
## 635517b9b32b3fa70f83    1.0211301 -0.004572989
## 64e6dc024b446b55739a    0.9965592 -0.004572989
## 6bcd49af023f477bbf37    2.3690506 -0.004572989
## 6bfa6a31a326dc706fdc    0.6248548 -0.004572989
## 6c5aa99681eb4d6905b1    3.7563649 -0.004572989
## 736bdf214487177022e5    0.8411342 -0.004572989
## 7543eafb309fd95fad6    1.3399003 -0.004572989
## 7fd7243a168e383f9e8b    3.2019562 -0.004572989
## 826e1d16e3b533fb245a    1.4995075 -0.004572989
## 83538a13d227275b86a5    1.0228485 -0.004572989
## 875b5d029de6eaeab7e7    1.8073601 -0.004572989
## 87b67b2c3ce58ae2ae1e    1.3057894 -0.004572989
## 8de5b549a2ea6dc6e183    0.7322179 -0.004572989
## 8f80bcc354d05f533ab4    5.9328611 -0.004572989
## 945fe99a248b31456606    6.6710195 -0.004572989
```

```
## 97d9363ed595442930a1 1.0216752 -0.004572989
## 98b0f6a6c10bbac3ed08 1.3050981 -0.004572989
## 9cc3e4db792b9a5f4e79 1.1140611 -0.004572989
## 9e57b44f8d0fde900aac 1.1058642 -0.004572989
## a07e6c06f538e78a6fc5 0.9027591 -0.004572989
## a4809ff054019f364d68 3.7129338 -0.004572989
## a99e74b610c40ecaa76f 1.3329477 -0.004572989
## b032e098afde0db1fe47 1.3556423 -0.004572989
## b6af4fba0707225b88c6 1.2939187 -0.004572989
## b759a10905ff6acf8c8b 0.9412904 -0.004572989
## b849ace4da81775ebd98 1.3179168 -0.004572989
## b97dd7e424b184fe9dd7 0.4913097 -0.004572989
## ba912daea85edd41c794 1.6797849 -0.004572989
## bb6bb1c0eb581072ba99 3.5077471 -0.004572989
## bd47d8131e7c18207855 2.7432081 -0.004572989
## c13597eae01d4a01c6b9 2.0089673 -0.004572989
## c25a28933629f381f987 2.4280837 -0.004572989
## c2e99bdb4d91daa8dda5 1.2583461 -0.004572989
## c309013be7be5bb5e89c 5.7846425 -0.004572989
## c34b33f5799854f89af7 0.9412746 -0.004572989
## cbc292b667a3879ecd4a 1.7098072 -0.004572989
## cc90d814885eaf54262f 0.9146298 -0.004572989
## d2d7e4c2d26d70b0994c 1.4310339 -0.004572989
## d34f86c865d6a3244c0d 0.8938314 -0.004572989
## db6df130100a85567a03 1.2310890 -0.004572989
## dc0a2924029e00ade90d 2.1047196 -0.004572989
## ddc74e6981d96523a1d4 0.6797167 -0.004572989
## ddd964c9902c136806eb 1.3476824 -0.004572989
## e1e8d661d37cf51a1b8c 0.7751443 -0.004572989
## e338df513c9d760904b9 1.2560155 -0.004572989
## eb5209c2c5d5ef9d8913 0.5015459 -0.004572989
## ed1fd295ccf29bd62cb1 1.6042984 -0.004572989
## ed98406eb2fb2cee3586 5.5610095 -0.004572989
## f1d2903ba93367937b48 1.4973794 -0.004572989
## f4146d4a6503f38cd7b4 1.1559739 -0.004572989
## f559a201d79bdf600dad 0.5175621 -0.004572989
## f705c2504ed873931a28 0.8285525 -0.004572989
## f7f047a44e75f386525c 1.1173004 -0.004572989
## f8d7b5b6a91e756d54ad 1.7967140 -0.004572989
## fa18c06cc119aa932a87 6.2161262 -0.004572989
##
## attr(,"class")
## [1] "coef.mer"
```

```
summary(Null.distr)
```

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method ['lmerModLmerTest']
## Formula: Ratio ~ 1 + stim_distance + (1 | subject)
## Data: DdataSW
##
## REML criterion at convergence: 20861.1
##
## Scaled residuals:
## Min 1Q Median 3Q Max
## -5.7817 -0.3608 -0.0694 0.2360 8.7722
```

```
##
## Random effects:
##   Groups   Name      Variance Std.Dev.
##  subject (Intercept) 2.901    1.703
##  Residual          1.107    1.052
## Number of obs: 6936, groups: subject, 86
##
## Fixed effects:
##               Estimate Std. Error      df t value Pr(>|t|)
## (Intercept)   1.994e+00  1.850e-01  8.538e+01  10.78   <2e-16 ***
## stim_distance -4.573e-03  2.032e-04  6.850e+03 -22.51   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##              (Intr)
## stim_distnc -0.095
#Convergence probs with stimuli distance as a random effect
#7 unique stim distance
unique(Ddata$stim_distance)

## [1] 15 25 50 75 100 150 200
Null.dist2r <- lmer(Ratio ~ 1 + stim_distance + (1 + stim_distance|subject), data = DdataSW)

## Warning in checkConv(attr(opt, "derivs"), opt$par, ctrl = control$checkConv, : Model failed to
## converge with max|grad| = 4.21628 (tol = 0.002, component 1)

## Warning in checkConv(attr(opt, "derivs"), opt$par, ctrl = control$checkConv, : Model is nearly under
## - Rescale variables?

coef(Null.dist2r)

## $subject
##               (Intercept) stim_distance
## 03b16361e0733153a514 0.22268739 8.562103e-04
## 03c20eaf3c5546c146e3 1.27506617 -4.554456e-03
## 07c208d8989c09cafdcb 3.59677236 -1.767025e-02
## 08c2c1e4288f21b62161 1.47663434 -7.248915e-03
## 0bc19d73dfcc79108774 6.36805719 -1.538257e-02
## 0d592ffa0ef1a0d4a525 1.44500713 1.391463e-02
## 0f2434b6a8611d7299c7 2.58675744 6.617636e-03
## 1153370ec05190a85eb0 0.47848946 3.316133e-04
## 19100d3830890f297f5c 1.41795010 -3.614676e-03
## 1a4a35d6d316c76b2554 0.31630270 4.460263e-04
## 1b12b81d3437180234b3 1.06910979 -3.962343e-03
## 1d3cd1067f2c7d38057d 8.44840836 -2.227341e-02
## 20616fb59a3d687092d0 1.00572185 -3.106410e-03
## 23f4a5027e9b70896a82 0.98248363 -4.199635e-03
## 292b6a1ff66453541947 0.63997270 -1.675796e-03
## 2ab98650a9465bb38516 1.15141130 -1.415975e-03
## 2c0ad1a3bae34df22e2d 1.90798297 -8.722697e-03
## 2c7549269a6dc4bf5c72 3.70469387 -1.623273e-02
## 2d6a9afc22d6a48697de 2.39171869 -2.721827e-03
## 38e4849b03aad5089e0e 1.03543019 -3.759571e-03
## 39cb033ba23e0a4204b4 0.94484329 -2.999279e-03
```

```

## 3a4cede6c4fc52ad4271 1.58447109 -8.093664e-03
## 3d85f7a3c49d9a89a36d 2.06359976 5.627543e-03
## 41956ce17cf84a1253c9 2.20168502 -1.065839e-02
## 499d33b349396d9f4919 1.18515643 -5.498623e-03
## 4d0ec1b598470d393c99 3.84043208 -9.589155e-03
## 575408d27d4f9024a6a3 0.08096101 -2.837079e-04
## 58024b3726839b0ad500 0.65729967 -2.470167e-03
## 584eecb0fbfc50073f14 1.96216341 -9.804411e-03
## 5b06e8406e2b0e210a97 8.35348086 -1.061304e-02
## 5ce421047cadb9563298 2.73210129 1.551593e-02
## 635517b9b32b3fa70f83 0.49699299 1.404737e-03
## 64e6dc024b446b55739a 0.97914981 -4.347656e-03
## 6bcd49af023f477bbf37 2.66658918 -7.963487e-03
## 6bfa6a31a326dc706fdc 0.19727113 3.200962e-04
## 6c5aa99681eb4d6905b1 2.21973696 1.267176e-02
## 736bdf214487177022e5 0.51022847 -7.838488e-04
## 7543eafb309fd95fad6 0.78636282 1.729223e-03
## 7fd7243a168e383f9e8b 3.05476234 -2.974679e-03
## 826e1d16e3b533fb245a 2.10799558 -1.140053e-02
## 83538a13d227275b86a5 0.90502642 -3.208392e-03
## 875b5d029de6eaeab7e7 2.18199178 -8.822356e-03
## 87b67b2c3ce58ae2ae1e 1.38860972 -5.494640e-03
## 8de5b549a2ea6dc6e183 0.66421664 -3.765542e-03
## 8f80bcc354d05f533ab4 9.25053079 -2.976478e-02
## 945fe99a248b31456606 7.63465315 -1.771544e-02
## 97d9363ed595442930a1 0.83241552 -2.397294e-03
## 98b0f6a6c10bbac3ed08 1.52334423 -7.032171e-03
## 9cc3e4db792b9a5f4e79 0.69720526 1.840103e-04
## 9e57b44f8d0fde900aac 0.60138442 1.179084e-03
## a07e6c06f538e78a6fc5 0.46281882 4.522693e-04
## a4809ff054019f364d68 6.38875352 -3.526315e-02
## a99e74b610c40ecaa76f 1.61552561 -7.763372e-03
## b032e098afde0db1fe47 1.52741002 -6.505957e-03
## b6af4fba0707225b88c6 1.26655537 -4.243328e-03
## b759a10905ff6acf8c8b 0.96865559 -4.854387e-03
## b849ace4da81775ebd98 0.85097998 7.466487e-04
## b97dd7e424b184fe9dd7 0.16486846 -8.556572e-04
## ba912daea85edd41c794 2.08486025 -9.164271e-03
## bb6bb1c0eb581072ba99 4.61395146 -1.600005e-02
## bd47d8131e7c18207855 4.30068722 -2.227909e-02
## c13597eae01d4a01c6b9 1.37226589 2.653880e-03
## c25a28933629f381f987 1.17508159 9.565496e-03
## c2e99bdb4d91daa8dda5 0.50384105 4.013282e-03
## c309013be7be5bb5e89c 8.03628896 -3.075512e-02
## c34b33f5799854f89af7 0.45018410 1.031876e-03
## cbc292b667a3879ecd4a 1.65938950 -3.993728e-03
## cc90d814885eaf54262f 0.60506427 -1.028281e-03
## d2d7e4c2d26d70b0994c 0.68105738 3.956823e-03
## d34f86c865d6a3244c0d 0.58621621 -1.049817e-03
## db6df130100a85567a03 1.23431070 -4.588739e-03
## dc0a2924029e00ade90d 2.20546128 -5.721443e-03
## ddc74e6981d96523a1d4 0.46821219 -2.134751e-03
## ddd964c9902c136806eb 1.69348222 -8.481589e-03
## e1e8d661d37cf51a1b8c 0.41796748 -4.836541e-04

```

```
## e338df513c9d760904b9 1.44504856 -6.699068e-03
## eb5209c2c5d5ef9d8913 0.09895318 3.996391e-05
## ed1fd295ccf29bd62cb1 2.39921986 -1.360267e-02
## ed98406eb2fb2cee3586 6.65569623 -1.777638e-02
## f1d2903ba93367937b48 1.73278673 -7.232626e-03
## f4146d4a6503f38cd7b4 0.92903809 -1.973462e-03
## f559a201d79bdf600dad 0.16779192 -5.602230e-04
## f705c2504ed873931a28 0.54023110 -1.266961e-03
## f7f047a44e75f386525c 1.14261491 -4.836246e-03
## f8d7b5b6a91e756d54ad 1.38848885 6.608490e-05
## fa18c06cc119aa932a87 7.51281350 -1.921480e-02
```

```
##
## attr(,"class")
## [1] "coef.mer"
```

```
fixef(Null.dist2r)
```

```
## (Intercept) stim_distance
## 2.036882451 -0.005084029
```

```
ranef(Null.dist2r)
```

```
## $subject
## (Intercept) stim_distance
## 03b16361e0733153a514 -1.81419506 0.0059402395
## 03c20eaf3c5546c146e3 -0.76181628 0.0005295730
## 07c208d8989c09cafdcb 1.55988991 -0.0125862215
## 08c2c1e4288f21b62161 -0.56024811 -0.0021648857
## 0bc19d73dfcc79108774 4.33117474 -0.0102985429
## 0d592ffa0ef1a0d4a525 -0.59187532 0.0189986577
## 0f2434b6a8611d7299c7 0.54987499 0.0117016656
## 1153370ec05190a85eb0 -1.55839299 0.0054156425
## 19100d3830890f297f5c -0.61893235 0.0014693528
## 1a4a35d6d316c76b2554 -1.72057975 0.0055300555
## 1b12b81d3437180234b3 -0.96777266 0.0011216866
## 1d3cd1067f2c7d38057d 6.41152591 -0.0171893825
## 20616fb59a3d687092d0 -1.03116060 0.0019776193
## 23f4a5027e9b70896a82 -1.05439882 0.0008843942
## 292b6a1ff66453541947 -1.39690975 0.0034082335
## 2ab98650a9465bb38516 -0.88547115 0.0036680547
## 2c0ad1a3bae34df22e2d -0.12889948 -0.0036386681
## 2c7549269a6dc4bf5c72 1.66781142 -0.0111487022
## 2d6a9afc22d6a48697de 0.35483624 0.0023622020
## 38e4849b03aad5089e0e -1.00145227 0.0013244583
## 39cb033ba23e0a4204b4 -1.09203917 0.0020847506
## 3a4ced6c4fc52ad4271 -0.45241136 -0.0030096350
## 3d85f7a3c49d9a89a36d 0.02671731 0.0107115721
## 41956ce17cf84a1253c9 0.16480257 -0.0055743600
## 499d33b349396d9f4919 -0.85172602 -0.0004145943
## 4d0ec1b598470d393c99 1.80354963 -0.0045051260
## 575408d27d4f9024a6a3 -1.95592144 0.0048003213
## 58024b3726839b0ad500 -1.37958278 0.0026138625
## 584eecb0fbfc50073f14 -0.07471904 -0.0047203819
## 5b06e8406e2b0e210a97 6.31659841 -0.0055290080
## 5ce421047caddb9563298 0.69521884 0.0205999562
## 635517b9b32b3fa70f83 -1.53988946 0.0064887664
```

```

## 64e6dc024b446b55739a -1.05773264 0.0007363736
## 6bcd49af023f477bbf37 0.62970673 -0.0028794581
## 6bfa6a31a326dc706fdc -1.83961132 0.0054041254
## 6c5aa99681eb4d6905b1 0.18285451 0.0177557875
## 736bdf214487177022e5 -1.52665398 0.0043001804
## 7543eafb309fd95fad6 -1.25051963 0.0068132521
## 7fd7243a168e383f9e8b 1.01787989 0.0021093500
## 826e1d16e3b533fb245a 0.07111313 -0.0063164964
## 83538a13d227275b86a5 -1.13185603 0.0018756368
## 875b5d029de6eaeab7e7 0.14510932 -0.0037383265
## 87b67b2c3ce58ae2ae1e -0.64827273 -0.0004106112
## 8de5b549a2ea6dc6e183 -1.37266581 0.0013184870
## 8f80bcc354d05f533ab4 7.21364834 -0.0246807548
## 945fe99a248b31456606 5.59777070 -0.0126314089
## 97d9363ed595442930a1 -1.20446693 0.0026867353
## 98b0f6a6c10bbac3ed08 -0.51353822 -0.0019481419
## 9cc3e4db792b9a5f4e79 -1.33967719 0.0052680395
## 9e57b44f8d0fde900aac -1.43549803 0.0062631135
## a07e6c06f538e78a6fc5 -1.57406363 0.0055362985
## a4809ff054019f364d68 4.35187107 -0.0301791184
## a99e74b610c40ecaa76f -0.42135684 -0.0026793423
## b032e098afde0db1fe47 -0.50947243 -0.0014219274
## b6af4fba0707225b88c6 -0.77032708 0.0008407008
## b759a10905ff6acf8c8b -1.06822686 0.0002296419
## b849ace4da81775ebd98 -1.18590247 0.0058306779
## b97dd7e424b184fe9dd7 -1.87201399 0.0042283720
## ba912daea85edd41c794 0.04797779 -0.0040802417
## bb6bb1c0eb581072ba99 2.57706901 -0.0109160168
## bd47d8131e7c18207855 2.26380477 -0.0171950637
## c13597eae01d4a01c6b9 -0.66461656 0.0077379095
## c25a28933629f381f987 -0.86180087 0.0146495251
## c2e99bdb4d91daa8dda5 -1.53304140 0.0090973110
## c309013be7be5bb5e89c 5.99940651 -0.0256710892
## c34b33f5799854f89af7 -1.58669836 0.0061159056
## cbc292b667a3879ecd4a -0.37749295 0.0010903017
## cc90d814885eaf54262f -1.43181818 0.0040557477
## d2d7e4c2d26d70b0994c -1.35582507 0.0090408526
## d34f86c865d6a3244c0d -1.45066624 0.0040342124
## db6df130100a85567a03 -0.80257175 0.0004952904
## dc0a2924029e00ade90d 0.16857883 -0.0006374133
## ddc74e6981d96523a1d4 -1.56867026 0.0029492781
## ddd964c9902c136806eb -0.34340023 -0.0033975603
## e1e8d661d37cf51a1b8c -1.61891497 0.0046003751
## e338df513c9d760904b9 -0.59183389 -0.0016150390
## eb5209c2c5d5ef9d8913 -1.93792927 0.0051239931
## ed1fd295ccf29bd62cb1 0.36233741 -0.0085186369
## ed98406eb2fb2cee3586 4.61881378 -0.0126923542
## f1d2903ba93367937b48 -0.30409572 -0.0021485965
## f4146d4a6503f38cd7b4 -1.10784437 0.0031105668
## f559a201d79bdf600dad -1.86909053 0.0045238062
## f705c2504ed873931a28 -1.49665135 0.0038170682
## f7f047a44e75f386525c -0.89426754 0.0002477834
## f8d7b5b6a91e756d54ad -0.64839361 0.0051501141
## fa18c06cc119aa932a87 5.47593105 -0.0141307720

```



```
##
## with conditional variances for "subject"
Null.condr <- lmer(Ratio ~ 1 + condition + (1|subject), data = DdataSW)
summary(Null.condr)

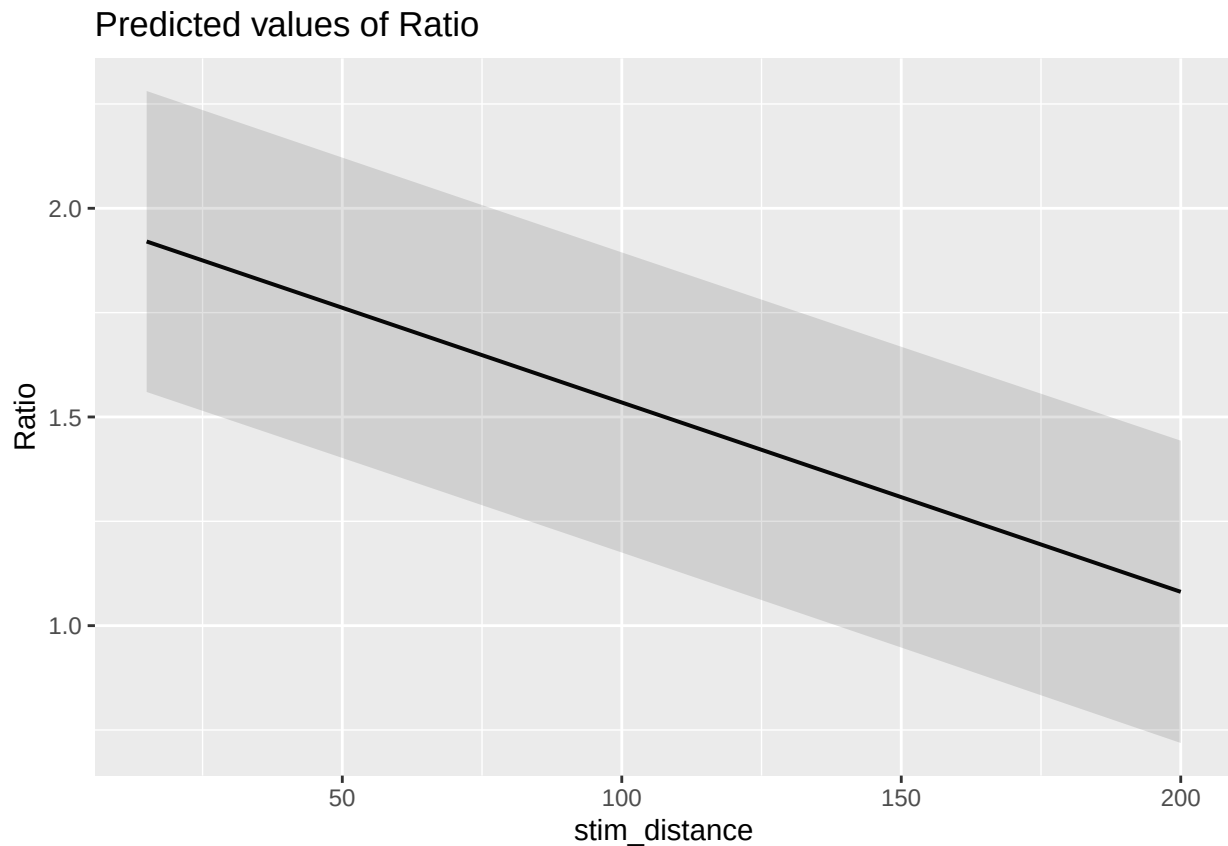
## Linear mixed model fit by REML. t-tests use Satterthwaite's method ['lmerModLmerTest']
## Formula: Ratio ~ 1 + condition + (1 | subject)
## Data: DdataSW
##
## REML criterion at convergence: 21333.9
##
## Scaled residuals:
## Min 1Q Median 3Q Max
## -5.8483 -0.3461 -0.0568 0.1766 8.7729
##
## Random effects:
## Groups Name Variance Std.Dev.
## subject (Intercept) 2.972 1.724
## Residual 1.188 1.090
## Number of obs: 6936, groups: subject, 86
##
## Fixed effects:
## Estimate Std. Error df t value Pr(>|t|)
## (Intercept) 2.2097 0.5938 82.7823 3.721 0.00036 ***
## condition -0.4048 0.3729 82.7574 -1.085 0.28092
## ---
## Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
## (Intr)
## condition -0.949

#Stim distance and condition
Modell1.dist.condr <- lmer(Ratio ~ 1 + stim_distance*condition + (1|subject), data = DdataSW)
summary(Modell1.dist.condr)

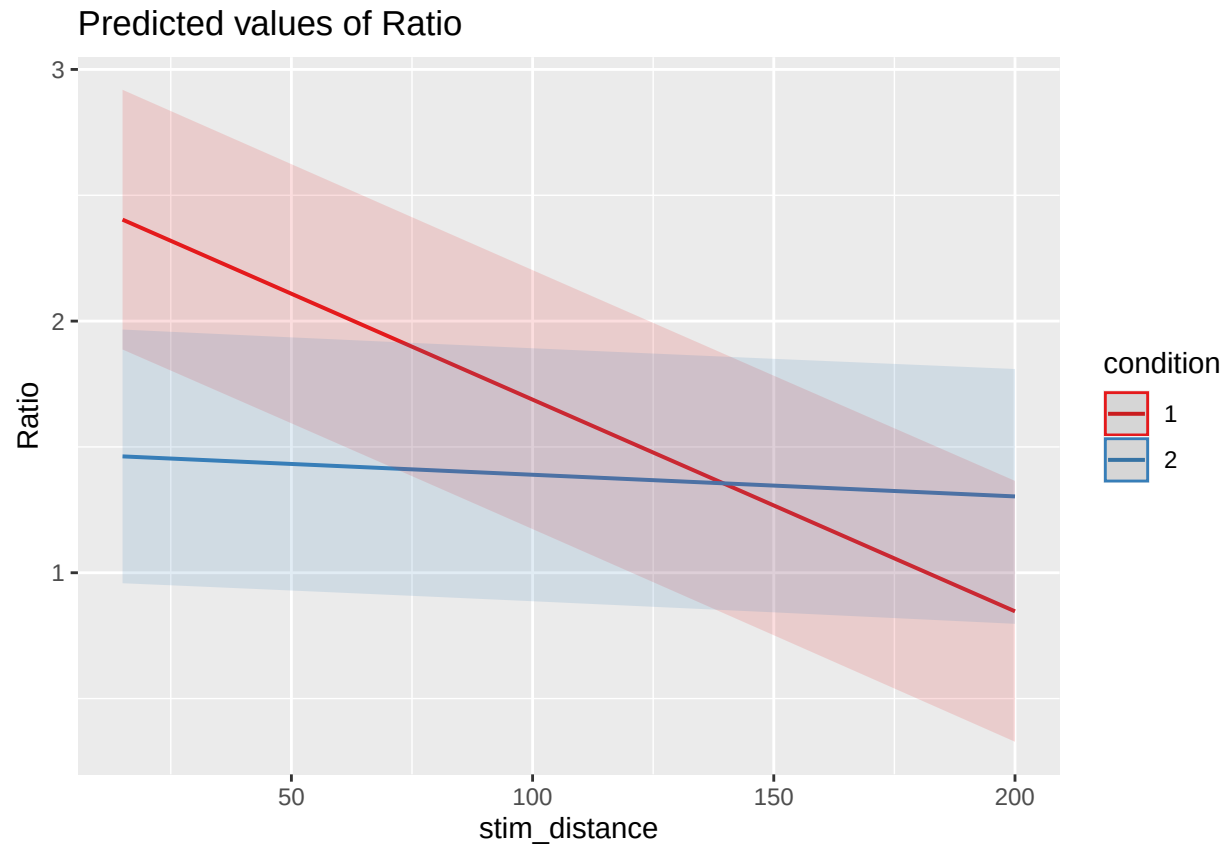
## Linear mixed model fit by REML. t-tests use Satterthwaite's method ['lmerModLmerTest']
## Formula: Ratio ~ 1 + stim_distance * condition + (1 | subject)
## Data: DdataSW
##
## REML criterion at convergence: 20519.4
##
## Scaled residuals:
## Min 1Q Median 3Q Max
## -5.7900 -0.3509 -0.0382 0.2273 9.2662
##
## Random effects:
## Groups Name Variance Std.Dev.
## subject (Intercept) 2.880 1.697
## Residual 1.051 1.025
## Number of obs: 6936, groups: subject, 86
##
## Fixed effects:
## Estimate Std. Error df t value Pr(>|t|)
```

```
## (Intercept)          3.583e+00  5.868e-01  8.443e+01   6.106   3e-08 ***
## stim_distance       -1.597e-02  6.294e-04  6.849e+03 -25.369 < 2e-16 ***
## condition          -1.054e+00  3.686e-01  8.442e+01  -2.860  0.00534 **
## stim_distance:condition  7.554e-03  3.960e-04  6.849e+03  19.073 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##      (Intr) stm_ds condtn
## stim_distnc -0.093
## condition   -0.949  0.088
## stm_dstnc:c  0.088 -0.949 -0.093
```

```
plot_model(Model1.dist.condr, type = "pred", terms = "stim_distance")
```



```
plot_model(Model1.dist.condr, type = "int", terms = "condition")
```



```
tab_model(Model1.dist.condr, show.df = TRUE)
```

Ratio
Predictors
Estimates
CI
p
df
(Intercept)
3.58
2.43 – 4.73
<0.001
6930.00
stim distance
-0.02
-0.02 – -0.01
<0.001

6930.00
condition
-1.05
-1.78 – -0.33
0.004
6930.00
stim distance * condition
0.01
0.01 – 0.01
<0.001
6930.00
Random Effects
2
1.05
00 subject
2.88
ICC
0.73
N subject
86
Observations
6936
Marginal R2 / Conditional R2
0.043 / 0.744

#Model comparison

anova(Null.distr, Null.condr, Model1.dist.condr)

refitting model(s) with ML (instead of REML)

Data: DdataSW

Models:

Null.distr: Ratio ~ 1 + stim_distance + (1 | subject)

Null.condr: Ratio ~ 1 + condition + (1 | subject)

Model1.dist.condr: Ratio ~ 1 + stim_distance * condition + (1 | subject)

##	npars	AIC	BIC	logLik	deviance	Chisq	Df	Pr(>Chisq)
## Null.distr	4	20852	20880	-10422	20844			
## Null.condr	4	21340	21368	-10666	21332	0.0	0	
## Model1.dist.condr	6	20501	20542	-10244	20489	843.6	2	< 2.2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Absolute error

```
#Sanity check: Stim distance and estimated distance with random int for participants
Null.dist.ae <- lmer(Ratio ~ 1 + stim_distance + (1|subject), data = DdataSW)
coef(Null.dist.ae)
```

```
## $subject
##               (Intercept) stim_distance
## 03b16361e0733153a514    0.6976788 -0.004572989
## 03c20eaf3c5546c146e3    1.2749770 -0.004572989
## 07c208d8989c09cafdcb    2.4330031 -0.004572989
## 08c2c1e4288f21b62161    1.2391279 -0.004572989
## 0bc19d73dfcc79108774    5.3263884 -0.004572989
## 0d592ffa0ef1a0d4a525    3.0261260 -0.004572989
## 0f2434b6a8611d7299c7    3.5096160 -0.004572989
## 1153370ec05190a85eb0    0.9078155 -0.004572989
## 19100d3830890f297f5c    1.5016416 -0.004572989
## 1a4a35d6d316c76b2554    0.7553138 -0.004572989
## 1b12b81d3437180234b3    1.1207766 -0.004572989
## 1d3cd1067f2c7d38057d    8.1731594 -0.004572989
## 20616fb59a3d687092d0    1.1328093 -0.004572989
## 23f4a5027e9b70896a82    1.0129727 -0.004572989
## 292b6a1ff66453541947    0.8924488 -0.004572989
## 2ab98650a9465bb38516    1.4281502 -0.004572989
## 2c0ad1a3bae34df22e2d    1.5414451 -0.004572989
## 2c7549269a6dc4bf5c72    2.6796081 -0.004572989
## 2d6a9afc22d6a48697de    2.5596642 -0.004572989
## 38e4849b03aad5089e0e    1.1049161 -0.004572989
## 39cb033ba23e0a4204b4    1.0812340 -0.004572989
## 3a4cede6c4fc52ad4271    1.2726463 -0.004572989
## 3d85f7a3c49d9a89a36d    2.9646767 -0.004572989
## 41956ce17cf84a1253c9    1.6649712 -0.004572989
## 499d33b349396d9f4919    1.1014596 -0.004572989
## 4d0ec1b598470d393c99    3.4023646 -0.004572989
## 575408d27d4f9024a6a3    0.4549252 -0.004572989
## 58024b3726839b0ad500    0.8396726 -0.004572989
## 584eecb0fbfc50073f14    1.5002433 -0.004572989
## 5b06e8406e2b0e210a97    8.2406454 -0.004572989
## 5ce421047caddb9563298    4.1420491 -0.004572989
## 635517b9b32b3fa70f83    1.0211301 -0.004572989
## 64e6dc024b446b55739a    0.9965592 -0.004572989
## 6bcd49af023f477bbf37    2.3690506 -0.004572989
## 6bfa6a31a326dc706fdc    0.6248548 -0.004572989
## 6c5aa99681eb4d6905b1    3.7563649 -0.004572989
## 736bdf214487177022e5    0.8411342 -0.004572989
## 7543eafb309fd95fad6    1.3399003 -0.004572989
## 7fd7243a168e383f9e8b    3.2019562 -0.004572989
## 826e1d16e3b533fb245a    1.4995075 -0.004572989
## 83538a13d227275b86a5    1.0228485 -0.004572989
## 875b5d029de6eaeab7e7    1.8073601 -0.004572989
## 87b67b2c3ce58ae2ae1e    1.3057894 -0.004572989
## 8de5b549a2ea6dc6e183    0.7322179 -0.004572989
## 8f80bcc354d05f533ab4    5.9328611 -0.004572989
## 945fe99a248b31456606    6.6710195 -0.004572989
```

```
## 97d9363ed595442930a1 1.0216752 -0.004572989
## 98b0f6a6c10bbac3ed08 1.3050981 -0.004572989
## 9cc3e4db792b9a5f4e79 1.1140611 -0.004572989
## 9e57b44f8d0fde900aac 1.1058642 -0.004572989
## a07e6c06f538e78a6fc5 0.9027591 -0.004572989
## a4809ff054019f364d68 3.7129338 -0.004572989
## a99e74b610c40ecaa76f 1.3329477 -0.004572989
## b032e098afde0db1fe47 1.3556423 -0.004572989
## b6af4fba0707225b88c6 1.2939187 -0.004572989
## b759a10905ff6acf8c8b 0.9412904 -0.004572989
## b849ace4da81775ebd98 1.3179168 -0.004572989
## b97dd7e424b184fe9dd7 0.4913097 -0.004572989
## ba912daea85edd41c794 1.6797849 -0.004572989
## bb6bb1c0eb581072ba99 3.5077471 -0.004572989
## bd47d8131e7c18207855 2.7432081 -0.004572989
## c13597eae01d4a01c6b9 2.0089673 -0.004572989
## c25a28933629f381f987 2.4280837 -0.004572989
## c2e99bdb4d91daa8dda5 1.2583461 -0.004572989
## c309013be7be5bb5e89c 5.7846425 -0.004572989
## c34b33f5799854f89af7 0.9412746 -0.004572989
## cbc292b667a3879ecd4a 1.7098072 -0.004572989
## cc90d814885eaf54262f 0.9146298 -0.004572989
## d2d7e4c2d26d70b0994c 1.4310339 -0.004572989
## d34f86c865d6a3244c0d 0.8938314 -0.004572989
## db6df130100a85567a03 1.2310890 -0.004572989
## dc0a2924029e00ade90d 2.1047196 -0.004572989
## ddc74e6981d96523a1d4 0.6797167 -0.004572989
## ddd964c9902c136806eb 1.3476824 -0.004572989
## e1e8d661d37cf51a1b8c 0.7751443 -0.004572989
## e338df513c9d760904b9 1.2560155 -0.004572989
## eb5209c2c5d5ef9d8913 0.5015459 -0.004572989
## ed1fd295ccf29bd62cb1 1.6042984 -0.004572989
## ed98406eb2fb2cee3586 5.5610095 -0.004572989
## f1d2903ba93367937b48 1.4973794 -0.004572989
## f4146d4a6503f38cd7b4 1.1559739 -0.004572989
## f559a201d79bdf600dad 0.5175621 -0.004572989
## f705c2504ed873931a28 0.8285525 -0.004572989
## f7f047a44e75f386525c 1.1173004 -0.004572989
## f8d7b5b6a91e756d54ad 1.7967140 -0.004572989
## fa18c06cc119aa932a87 6.2161262 -0.004572989
##
## attr(,"class")
## [1] "coef.mer"
```

```
summary(Null.dist.ae)
```

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method ['lmerModLmerTest']
## Formula: Ratio ~ 1 + stim_distance + (1 | subject)
## Data: DdataSW
##
## REML criterion at convergence: 20861.1
##
## Scaled residuals:
## Min 1Q Median 3Q Max
## -5.7817 -0.3608 -0.0694 0.2360 8.7722
```

```
##
## Random effects:
##   Groups   Name                Variance Std.Dev.
##  subject (Intercept) 2.901      1.703
##  Residual          1.107      1.052
## Number of obs: 6936, groups:  subject, 86
##
## Fixed effects:
##              Estimate Std. Error      df t value Pr(>|t|)
## (Intercept)   1.994e+00  1.850e-01  8.538e+01  10.78   <2e-16 ***
## stim_distance -4.573e-03  2.032e-04  6.850e+03 -22.51   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##              (Intr)
## stim_distnc -0.095
#Convergence probs with stimuli distance as a random effect
#7 unique stim distance
unique(Ddata$stim_distance)

## [1] 15 25 50 75 100 150 200
Null.dist2.ae <- lmer(AbsoluteError ~ 1 + stim_distance + (1 + stim_distance|subject), data = DdataSW)

## Warning in checkConv(attr(opt, "derivs"), opt$par, ctrl = control$checkConv, : Model failed to
## converge with max|grad| = 0.622212 (tol = 0.002, component 1)
coef(Null.dist2.ae)

## $subject
##              (Intercept) stim_distance
## 03b16361e0733153a514    5.7943577    0.59773924
## 03c20eaf3c5546c146e3   -12.6474689    0.55100176
## 07c208d8989c09cafdcb    54.4513736    0.01737108
## 08c2c1e4288f21b62161   -14.5931975    0.75947233
## 0bc19d73dfcc79108774   173.4971276    1.47064528
## 0d592ffa0ef1a0d4a525  -115.8859966    3.66770528
## 0f2434b6a8611d7299c7   -49.2280046    3.38854104
## 1153370ec05190a85eb0    -3.4441484    0.51662133
## 19100d3830890f297f5c    16.4170177    0.39662894
## 1a4a35d6d316c76b2554     2.2472742    0.59952732
## 1b12b81d3437180234b3   -12.5359196    0.69351719
## 1d3cd1067f2c7d38057d    92.5523571    1.63228832
## 20616fb59a3d687092d0   -11.5934271    0.58528364
## 23f4a5027e9b70896a82   -11.1812170    0.73255885
## 292b6a1ff66453541947    -2.7069658    0.63761507
## 2ab98650a9465bb38516     0.4605557    0.10562135
## 2c0ad1a3bae34df22e2d    -4.9862819    0.50464094
## 2c7549269a6dc4bf5c72    79.3934415   -0.06978172
## 2d6a9afc22d6a48697de    14.7543962    0.90951776
## 38e4849b03aad5089e0e   -11.9419308    0.62415608
## 39cb033ba23e0a4204b4   -19.3820885    0.68585669
## 3a4cede6c4fc52ad4271    -6.1924393    0.68161861
## 3d85f7a3c49d9a89a36d   -46.8359071    2.38910868
```

## 41956ce17cf84a1253c9	15.5126790	0.38739606
## 499d33b349396d9f4919	-12.9902103	0.76818279
## 4d0ec1b598470d393c99	78.2354280	0.67838002
## 575408d27d4f9024a6a3	-0.4568031	0.96685103
## 58024b3726839b0ad500	-12.7350614	0.82375129
## 584eecb0fbfc50073f14	4.9572291	0.55980722
## 5b06e8406e2b0e210a97	26.2018537	6.44813657
## 5ce421047cadb9563298	-57.9629944	4.29627557
## 635517b9b32b3fa70f83	5.6758751	0.25342336
## 64e6dc024b446b55739a	-18.2076604	0.83866836
## 6bcd49af023f477bbf37	26.9875423	0.31693843
## 6bfa6a31a326dc706fdc	3.3481519	0.72407105
## 6c5aa99681eb4d6905b1	-77.5450268	3.84454798
## 736bdf214487177022e5	0.5688173	0.59520207
## 7543eafb3a309fd95fad6	1.5418782	0.16336668
## 7fd7243a168e383f9e8b	27.5432383	1.59807789
## 826e1d16e3b533fb245a	7.0866207	0.63310722
## 83538a13d227275b86a5	-15.7823387	0.73044854
## 875b5d029de6eaeab7e7	14.3486957	0.18937684
## 87b67b2c3ce58ae2ae1e	-11.6023687	0.57896820
## 8de5b549a2ea6dc6e183	-6.9466059	0.94331623
## 8f80bcc354d05f533ab4	463.5983605	0.22194932
## 945fe99a248b31456606	79.4414885	3.55939835
## 97d9363ed595442930a1	-11.6336908	0.64158359
## 98b0f6a6c10bbac3ed08	-6.0082571	0.61844304
## 9cc3e4db792b9a5f4e79	-1.6130453	0.29633061
## 9e57b44f8d0fde900aac	2.4683479	0.22085554
## a07e6c06f538e78a6fc5	3.4775015	0.43676151
## a4809ff054019f364d68	120.7940791	-0.46485714
## a99e74b610c40ecaa76f	-3.7831546	0.58685166
## b032e098afde0db1fe47	-0.8883587	0.49843613
## b6af4fba0707225b88c6	-12.2078915	0.53865290
## b759a10905ff6acf8c8b	-10.0910949	0.83780962
## b849ace4da81775ebd98	4.5852588	0.09220431
## b97dd7e424b184fe9dd7	-3.5167019	0.99726547
## ba912daea85edd41c794	6.5208822	0.39633472
## bb6bb1c0eb581072ba99	116.8846375	0.27340530
## bd47d8131e7c18207855	66.7054379	-0.07237540
## c13597eae01d4a01c6b9	-12.7965558	0.90906410
## c25a28933629f381f987	-37.5875992	1.94945711
## c2e99bdb4d91daa8dda5	11.3624095	0.10945665
## c309013be7be5bb5e89c	121.7462015	1.65517420
## c34b33f5799854f89af7	12.7365988	0.28380940
## cbc292b667a3879ecd4a	13.8096815	0.11986560
## cc90d814885eaf54262f	-9.5268073	0.64860873
## d2d7e4c2d26d70b0994c	-2.7565842	0.51469979
## d34f86c865d6a3244c0d	-5.3212945	0.61666379
## db6df130100a85567a03	-16.2334559	0.61321740
## dc0a2924029e00ade90d	33.3265854	0.22137620
## ddc74e6981d96523a1d4	6.9993606	0.78993202
## ddd964c9902c136806eb	-4.7564802	0.68709796
## e1e8d661d37cf51a1b8c	-0.8404017	0.65833438
## e338df513c9d760904b9	-11.4703307	0.68238933
## eb5209c2c5d5ef9d8913	1.0284262	0.88781298


```
## ed1fd295ccf29bd62cb1      4.7480477      0.64894669
## ed98406eb2fb2cee3586      87.1651001      2.23640869
## f1d2903ba93367937b48     -1.5584822      0.44559203
## f4146d4a6503f38cd7b4     -9.7130186      0.45847604
## f559a201d79bdf600dad     -1.7075545      0.93538659
## f705c2504ed873931a28     -8.6547970      0.73068729
## f7f047a44e75f386525c    -17.3362292      0.75461900
## f8d7b5b6a91e756d54ad    -13.0712264      0.71562859
## fa18c06cc119aa932a87    144.2333380      2.31601316
##
## attr(,"class")
## [1] "coef.mer"
```

```
fixef(Null.dist2.ae)
```

```
##      (Intercept) stim_distance
##      14.1017509      0.9149223
```

```
ranef(Null.dist2.ae)
```

```
## $subject
##      (Intercept) stim_distance
## 03b16361e0733153a514    -8.3073932    -0.317183039
## 03c20eaf3c5546c146e3   -26.7492198    -0.363920517
## 07c208d8989c09cafdcb    40.3496227    -0.897551197
## 08c2c1e4288f21b62161   -28.6949484    -0.155449949
## 0bc19d73dfcc79108774   159.3953766      0.555723004
## 0d592ffa0ef1a0d4a525  -129.9877475      2.752783002
## 0f2434b6a8611d7299c7   -63.3297555      2.473618765
## 1153370ec05190a85eb0   -17.5458993    -0.398300942
## 19100d3830890f297f5c     2.3152668    -0.518293340
## 1a4a35d6d316c76b2554   -11.8544767    -0.315394960
## 1b12b81d3437180234b3   -26.6376705    -0.221405083
## 1d3cd1067f2c7d38057d    78.4506062      0.717366041
## 20616fb59a3d687092d0   -25.6951780    -0.329638631
## 23f4a5027e9b70896a82   -25.2829679    -0.182363423
## 292b6a1ff66453541947   -16.8087167    -0.277307208
## 2ab98650a9465bb38516   -13.6411953    -0.809300925
## 2c0ad1a3bae34df22e2d   -19.0880328    -0.410281334
## 2c7549269a6dc4bf5c72    65.2916905    -0.984703993
## 2d6a9afc22d6a48697de     0.6526452    -0.005404521
## 38e4849b03aad5089e0e   -26.0436817    -0.290766195
## 39cb033ba23e0a4204b4   -33.4838395    -0.229065582
## 3a4ced6c4fc52ad4271   -20.2941902    -0.233303666
## 3d85f7a3c49d9a89a36d   -60.9376580      1.474186400
## 41956ce17cf84a1253c9     1.4109281    -0.527526214
## 499d33b349396d9f4919   -27.0919612    -0.146739485
## 4d0ec1b598470d393c99    64.1336771    -0.236542258
## 575408d27d4f9024a6a3   -14.5585540      0.051928753
## 58024b3726839b0ad500   -26.8368123    -0.091170985
## 584eecb0fbfc50073f14    -9.1445219    -0.355115059
## 5b06e8406e2b0e210a97    12.1001028      5.533214299
## 5ce421047cadb9563298   -72.0647453      3.381353298
## 635517b9b32b3fa70f83    -8.4258759    -0.661498912
## 64e6dc024b446b55739a   -32.3094113    -0.076253912
## 6bcd49af023f477bbf37    12.8857914    -0.597983850
```

```

## 6bfa6a31a326dc706fdc -10.7535990 -0.190851226
## 6c5aa99681eb4d6905b1 -91.6467777 2.929625707
## 736bdf214487177022e5 -13.5329336 -0.319720209
## 7543eafba309fd95fad6 -12.5598727 -0.751555599
## 7fd7243a168e383f9e8b 13.4414874 0.683155616
## 826e1d16e3b533fb245a -7.0151302 -0.281815054
## 83538a13d227275b86a5 -29.8840897 -0.184473738
## 875b5d029de6eaeab7e7 0.2469447 -0.725545438
## 87b67b2c3ce58ae2ae1e -25.7041196 -0.335954078
## 8de5b549a2ea6dc6e183 -21.0483568 0.028393955
## 8f80bcc354d05f533ab4 449.4966096 -0.692972961
## 945fe99a248b31456606 65.3397376 2.644476073
## 97d9363ed595442930a1 -25.7354417 -0.273338689
## 98b0f6a6c10bbac3ed08 -20.1100080 -0.296479234
## 9cc3e4db792b9a5f4e79 -15.7147962 -0.618591668
## 9e57b44f8d0fde900aac -11.6334030 -0.694066732
## a07e6c06f538e78a6fc5 -10.6242494 -0.478160765
## a4809ff054019f364d68 106.6923282 -1.379779414
## a99e74b610c40ecaa76f -17.8849055 -0.328070616
## b032e098afde0db1fe47 -14.9901096 -0.416486142
## b6af4fba0707225b88c6 -26.3096424 -0.376269373
## b759a10905ff6acf8c8b -24.1928458 -0.077112651
## b849ace4da81775ebd98 -9.5164921 -0.822717964
## b97dd7e424b184fe9dd7 -17.6184528 0.082343191
## ba912daea85edd41c794 -7.5808687 -0.518587561
## bb6bb1c0eb581072ba99 102.7828866 -0.641516975
## bd47d8131e7c18207855 52.6036870 -0.987297679
## c13597eae01d4a01c6b9 -26.8983067 -0.005858174
## c25a28933629f381f987 -51.6893501 1.034534835
## c2e99bdb4d91daa8dda5 -2.7393414 -0.805465621
## c309013be7be5bb5e89c 107.6444506 0.740251927
## c34b33f5799854f89af7 -1.3651521 -0.631112880
## cbc292b667a3879ecd4a -0.2920694 -0.795056673
## cc90d814885eaf54262f -23.6285583 -0.266313545
## d2d7e4c2d26d70b0994c -16.8583351 -0.400222486
## d34f86c865d6a3244c0d -19.4230455 -0.298258486
## db6df130100a85567a03 -30.3352068 -0.301704874
## dc0a2924029e00ade90d 19.2248345 -0.693546071
## ddc74e6981d96523a1d4 -7.1023903 -0.124990259
## ddd964c9902c136806eb -18.8582311 -0.227824316
## e1e8d661d37cf51a1b8c -14.9421526 -0.256587900
## e338df513c9d760904b9 -25.5720816 -0.232532943
## eb5209c2c5d5ef9d8913 -13.0733247 -0.027109295
## ed1fd295ccf29bd62cb1 -9.3537032 -0.265975582
## ed98406eb2fb2cee3586 73.0633492 1.321486412
## f1d2903ba93367937b48 -15.6602331 -0.469330249
## f4146d4a6503f38cd7b4 -23.8147695 -0.456446232
## f559a201d79bdf600dad -15.8093054 0.020464310
## f705c2504ed873931a28 -22.7565479 -0.184234983
## f7f047a44e75f386525c -31.4379801 -0.160303272
## f8d7b5b6a91e756d54ad -27.1729773 -0.199293687
## fa18c06cc119aa932a87 130.1315871 1.401090889
##
## with conditional variances for "subject"

```

```
Null.cond.ae <- lmer(AbsoluteError ~ 1 + condition + (1|subject), data = DdataSW)
summary(Null.cond.ae)
```

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method ['lmerModLmerTest']
## Formula: AbsoluteError ~ 1 + condition + (1 | subject)
## Data: DdataSW
##
## REML criterion at convergence: 82131.2
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -3.5776 -0.4342 -0.1424  0.3383  7.6467
##
## Random effects:
## Groups Name Variance Std.Dev.
## subject (Intercept) 7739 87.97
## Residual 7719 87.86
## Number of obs: 6936, groups: subject, 86
##
## Fixed effects:
## Estimate Std. Error df t value Pr(>|t|)
## (Intercept) 85.762 30.424 83.625 2.819 0.00601 **
## condition 0.748 19.105 83.570 0.039 0.96886
## ---
## Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
## (Intr)
## condition -0.949
```

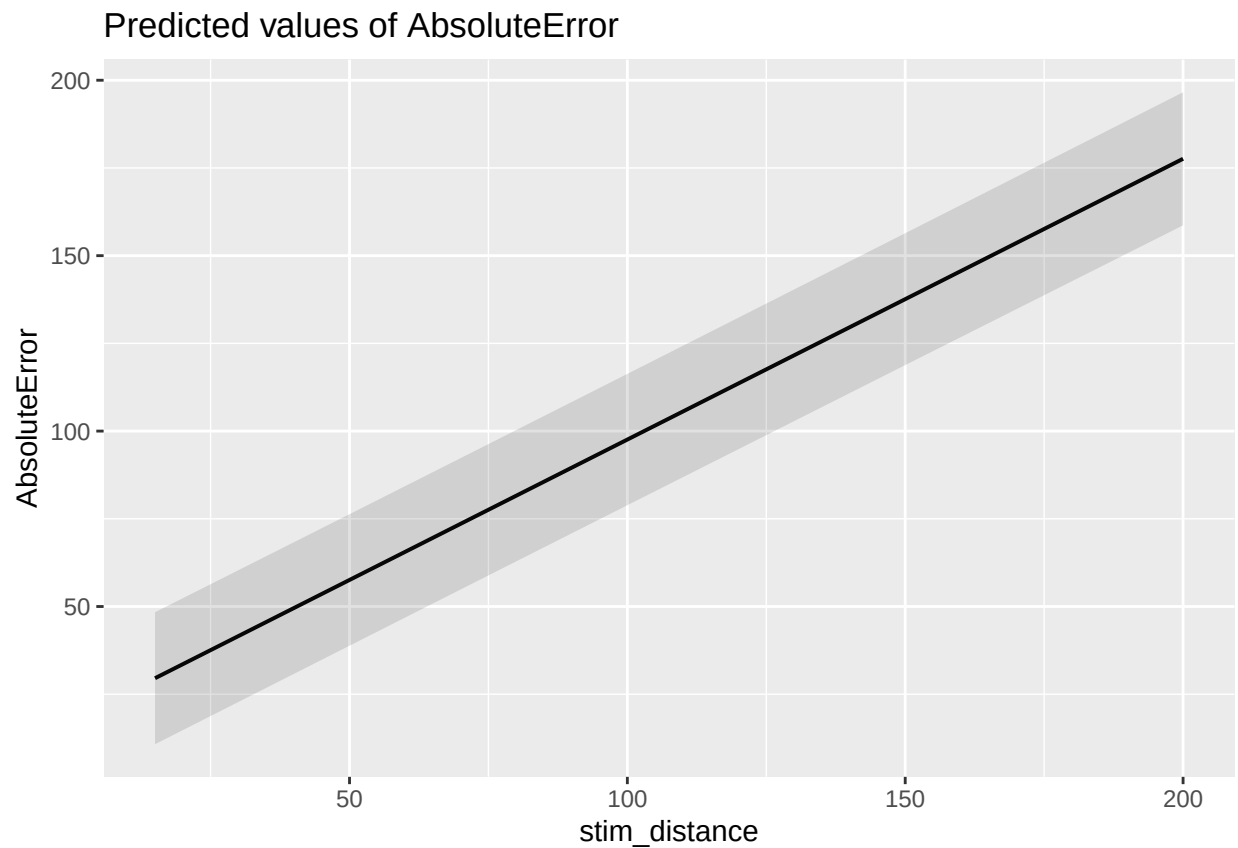
#Stim distance and condition

```
Model1.dist.cond.ae <- lmer(AbsoluteError ~ 1 + stim_distance*condition + (1|subject), data = DdataSW)
summary(Model1.dist.cond.ae)
```

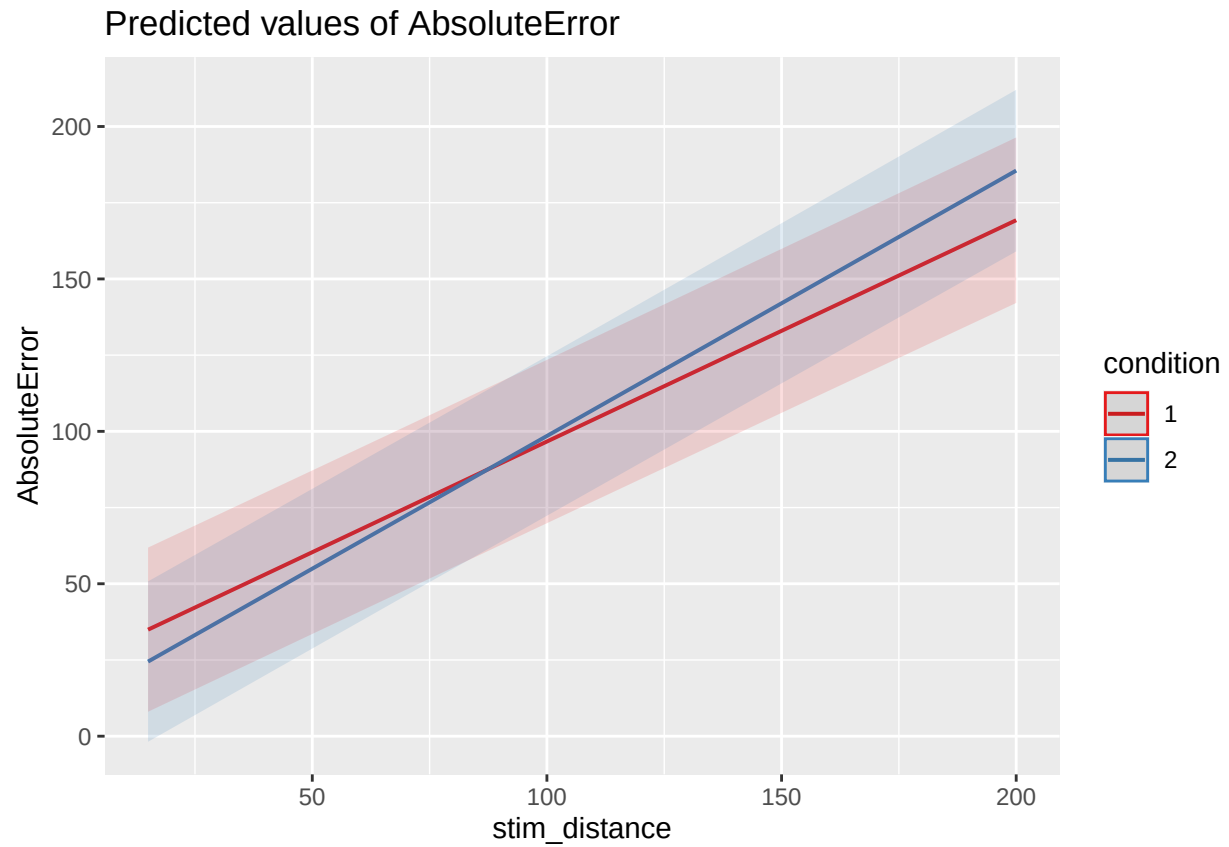
```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method ['lmerModLmerTest']
## Formula: AbsoluteError ~ 1 + stim_distance * condition + (1 | subject)
## Data: DdataSW
##
## REML criterion at convergence: 79431
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -5.3716 -0.2939 -0.0101  0.2435  7.9262
##
## Random effects:
## Groups Name Variance Std.Dev.
## subject (Intercept) 7771 88.15
## Residual 5196 72.09
## Number of obs: 6936, groups: subject, 86
##
## Fixed effects:
## Estimate Std. Error df t value Pr(>|t|)
## (Intercept) 36.68224 30.65758 86.31208 1.197 0.235
## stim_distance 0.58154 0.04425 6850.98928 13.143 < 2e-16 ***
```

```
## condition          -12.63664   19.25496   86.30098  -0.656    0.513
## stim_distance:condition    0.14455    0.02784 6851.24756   5.191 2.15e-07 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##          (Intr) stm_ds condtn
## stim_distnc -0.125
## condition   -0.949  0.119
## stm_dstnc:c  0.118 -0.949 -0.125
```

```
plot_model(Model11.dist.cond.ae, type = "pred", terms = "stim_distance")
```



```
plot_model(Model11.dist.cond.ae, type = "int", terms = "condition")
```



```
tab_model(Model1.dist.cond.ae, show.df = TRUE)
```

AbsoluteError

Predictors

Estimates

CI

p

df

(Intercept)

36.68

-23.42 – 96.78

0.232

6930.00

stim distance

0.58

0.49 – 0.67

<0.001

6930.00
condition
-12.64
-50.38 – 25.11
0.512
6930.00
stim distance * condition
0.14
0.09 – 0.20
<0.001
6930.00
Random Effects
2
5196.47
00 subject
7771.23
ICC
0.60
N subject
86
Observations
6936
Marginal R2 / Conditional R2
0.163 / 0.665

#Model comparison

anova(Null.dist.ae, Null.cond.ae, Model1.dist.cond.ae)

refitting model(s) with ML (instead of REML)

Data: DdataSW

Models:

Null.dist.ae: Ratio ~ 1 + stim_distance + (1 | subject)

Null.cond.ae: AbsoluteError ~ 1 + condition + (1 | subject)

Model1.dist.cond.ae: AbsoluteError ~ 1 + stim_distance * condition + (1 | subject)

	npars	AIC	BIC	logLik	deviance	Chisq	Df	Pr(>Chisq)
## Null.dist.ae	4	20852	20880	-10422	20844			
## Null.cond.ae	4	82153	82181	-41073	82145	0.0	0	
## Model1.dist.cond.ae	6	79445	79486	-39716	79433	2712.3	2	< 2.2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1